P-B-172B
March 16, 1973
SUPERSEDING
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May 15, 1965

FEDERAL SPECIFICATION

BELTS AND BANDS, ABRASIVE (COATED CLOTH, POWER MACHINE)

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers coated abrasive cloth belts and bands used with power machines for sanding and finishing wood, metal and other materials.

1.2 Classification.

1.2.1 Types, classes, and compositions. The belt and bands shall be of the following types, classes, and compositions as specified (see 6.2):

Type I - Belts.

Class 1 - Glue bond.
Composition:
1 - Aluminum oxide, closed coat, drills.
2 - Aluminum oxide, open coat, drills.
3 - Silicon carbide, closed coat, drills.
4 - Garnet, closed coat, jeans.
5 - Garnet, closed coat, drills.
6 - Garnet, open coat, jeans.
7 - Garnet, open coat, drills.

Class 2 - Resin over glue bond.
Composition:
1 - Aluminum oxide, closed coat, drills.
2 - Aluminum oxide, open coat, drills.
3 - Silicon carbide, closed coat, drills.
4 - Garnet, closed coat, drills.
6 - Garnet, open coat, jeans.

Class 3 - Resin bond.
Composition:
1 - Aluminum oxide, closed coat, drills.
2 - Aluminum oxide, open coat, drills.
3 - Silicon carbide, closed coat, drills.

Class 4 - Waterproof (wet sanding).
Composition:
1 - Aluminum oxide, closed coat, drills.
3 - Silicon carbide, closed coat, drills.

Type II - Bands

Class 1 - Glue bond.
Composition:
1 - Aluminum oxide, closed coat, drills.
2 - Aluminum oxide, open coat, drills.
5 - Garnet, closed coat, drills.

FSC 5345

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Class 2 - Resin over glue bond.
Composition:
1 - Aluminum oxide, closed coat, drills.
2 - Aluminum oxide, open coat, drills.
5 - Garnet, closed coat, drills.

Class 3 - Resin bond.
Composition:
1 - Aluminum oxide, closed coat, drills.
2 - Aluminum oxide, open coat, drills.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking of Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-3816 - Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Commerce Product Standard:

PS 8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Coated abrasive cloth belts and bands shall be fabricated from cloth coated on one side with silicon carbide, aluminum oxide, or garnet abrasive as hereinafter specified.

3.2 Back cloth. Jeans ("J") backing cloth and drills ("X") shall be made from cotton grey cloth conforming to the requirements of Table I.

TABLE I - Grey cloth backing requirements

Types	Cloth Type	Yards(1) per pound		Thread Count Per Inch			
				Warp		Fill	
		Min.	Max.	Min.	Max.	Min.	Max.
I	Jeans	3.36	3.72	93	101	61	69
I and II	Drills(2)	2.64	2.92	73	81	45	53
II	Drills(2)	2.64	2.92	69	77	45	53

(1) Nominal yards per pound of grey cloth is on a 30-inch wide basis at 70 degrees F. ± 2 Degrees F. and 65 percent plus or minus 2 percent relative humidity.
(2) Drills cloth of the above requirements is suitable for innerliners of Type II bands.

3.2.1 Nonwaterproof (classes 1, 2, and 3). The cloth backing used for nonwaterproof coated abrasive belts and bands shall consist of cotton grey cloth as specified in 3.2 and shall be dyed and properly filled prior to the application of the abrasive coating, to produce a satisfactory nonwaterproof backing.

3.2.2 Waterproof (Class 4). The cloth backing used for waterproof abrasive belts shall consist of a cotton cloth as specified in 3.2 and shall be dyed and properly filled with waterproof materials prior to the application of the abrasive coating to produce a satisfactory waterproof backing.

3.3 Adhesive bond.

3.3.1 Class 1, glue bond. The making coat of adhesive and the sizing coat shall consist of a good quality cattle-hide glue. Either or both adhesive coats may contain modifying fillers.

3.3.2 Class 2, resin over glue bond. The making coat of adhesive shall consist of a good quality cattle-hide glue and the sizing coat shall consist of a heat hardenable resin. Either or both adhesive coats may contain modifying fillers.

3.3.3 Class 3, resin over resin and Class 4 water bonds. The making coat of adhesive and the sizing coat shall consist of a heat hardenable resin. Either or both adhesive coats may contain modifying fillers.

3.4 Abrasive grain. The abrasive grain shall not contain more than 5 percent extraneous material and shall be furnished in grit sizes in conformance with Table II, as specified (see 6.2).

3.4.1 Aluminum oxide. The aluminum oxide abrasive grain used for the products specified herein shall be crushed from high quality, fused aluminum oxide.

3.4.2 Silicon carbide. The silicon carbide abrasive grain used for the products specified herein shall be crushed from high quality, silicon carbide.

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3.4.3 Garnet. The garnet abrasive grain used for the products specified herein shall be crushed from clean, hard garnet of abrasive quality.

3.4.4 Grit size.

3.4.4.1 Control of grit sizes. The grading, control, and tolerances of the abrasive grain on the coated product shall be in accordance with PS-8.

3.5 Coating.

3.5.1 Method of coating. The coated abrasive cloths from which belts and bands are formed shall be coated on one side of the cloth backing with abrasive grain applied in such a manner that a sharp, fast-cutting coated abrasive product results.

3.5.2 Types of coating.

3.5.2.1 Closed coat. Closed coat abrasive cloths for belts and bands shall have the abrasive grain evenly deposited to provide complete coverage of the backing surface.

3.5.2.2 Open coat. Open coat abrasive cloth for belts and bands shall have not less than 50 percent nor more than 70 percent coverage of the backing surface with abrasive grain. The abrasive grain deposit shall be rigidly controlled to provide uniform open space between the abrasive grains.

TABLE II. Abrasive grit sizes for types I and II, classes 1, 2, 3, and 4 belts and bands

TYPE	CLASS	COMPOSITION	GRIT SIZE														
WEIGHT BACK - (Thicker and Less Flexible Than J Weight)																	
and II	1	1	400	320	280	240	220	180	150	120	100	80	60	50	40	36	24
and II	1	2								120	100	80	60	50	40	36	24
	1	3		320		240	220	180	150	120	100	80	60	50	40	36	24
and II	1	5							150	120	100	80	60	50	40	36	24
	1	7								120	100	80	60	50	40	36	24
and II	2	1	400	320	280	240	220	180	150	120	100	80	60	50	40	36	24
and II	2	2								120	100	80	60	50	40	36	24
	2	3	400	320	280	240	220	180	150	120	100	80	60	50	40	36	24
and II	2	5							150	120	100	80	60	50	40	361	24
and II	3	1	400	320	280	240	220	180	150	120	100	80	60	50	40	36	24
and II	3	2					220	180	150	120	100	80	60	50	40	36	24
	3	3	400	320	280	240	220	180	150	120	100	80	60	50	40	36	24
	4	1	400	320		240	220	180	150	120	100	80	60	50	40	36	24
	4	3	400	320	280	240	220	180	150	120	100	80	60	50	40	36	24
WEIGHT BACK - (Thinner and More Flexible than X Weight)																	
	1	4							150	120	100	80	60				
	1	6							150	120	100	80	60				
	2	4							150	120	100	80	60				
	2	6							150	120	100	80	60				

NOTE: Where Class 2 is shown, it is permissible to fill the requirements for these items with Class 3 material.

3.6 Type I belts. Type I belts shall consist of a single layer of abrasive coated cloth neatly spliced to form a continuous belt.

3.6.1 Splices. Coated abrasive belts shall be smoothly spliced by use of a lap type joint or a reinforced butt joint in accordance with industry practice. The two ends of the coated abrasive cloth shall be joined with a suitable adhesive to prevent separation of the splice during normal working conditions.

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3.6.2 Marking for identification. Coated abrasive belts shall be clearly marked as follows except as indicated in Table II:

(a) Kind of abrasive grain (either generic name or trade name of abrasive grain or product).
(b) Grit size as shown in Table II.
(c) Backing weight "X" for drills and "J" for jeans, except for class 4 products which are all coated on drills but are not necessarily marked with the letter "X".
(d) All "open" coat belts shall be marked "open".
(e) Belt size.
(f) Directional arrow.
(g) Manufacturer's name or trademark of such known character than the source of manufacture may be readily determined.
(h) Specification Number, Type, and Class.

3.6.3 The belts when conditioned for at least 24 hours at 70 degrees ± 2 degrees F. and 50 percent plus or minus 2 percent relative humidity, shall conform to the dimensions shown in table III for the size specified (see 6.2).

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TABLE III. Type I belts, Dimensions

Width	Length (total inside measurement)
Tolerances:	Tolerances:
Under 6 inches: ± 1/16	Under 85 inches: ± 1/8
6 inches and over:	85 to 150 inches: ± 1/4
40 grit size and finer: ± 1/16	150 inches and over: ± 3/8
36 grit size and coarser: ± 3/32	

Width	Length
4	6-1/2
1	12
3-1/2	15-1/2
3	18
1, 1-1/2, 2	18-5/16
2	20
3/4	20-1/2
2, 2-1/2, 3, 4	21
2-1/4	21-5/16
8	21-5/8
1, 1-1/2	22
3	23-3/4
3, 4	24
3-1/2, 4-1/2	26
3, 4	27
4	35
4	36
4, 6	36-1/4
2	38-3/8
1	42
2, 3, 4	44-3/16
1, 2, 2-1/2, 6	48
4, 6	48-5/16
4	52-1/2
4	53
4	54
6	55
1-1/2, 2, 2-1/2, 4,6	60
4	64
4	64-1/2
10	70-1/2
1-1/2, 2, 2-1/2	72
1, 1-1/4, 1-1/2, 1-3/4	77
4	82
4	84
9	85
4	106
8,9	107
1	114
4	118
2, 3	120
2, 3, 4, 6	132
14	133
3	141
1	144
20	145
8	150
1	159
3, 4, 6	168
1, 6	171
7	180

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3.6.4 Class 1, glue bond. Class 1 belts shall have a making coat of glue (see 3.3.1) applied to the backing, and after the abrasive grain is deposited on the Making coat, a sizing coat of glue shall be applied to complete the bond.

3.6.5 Class 2, resin over glue bond. Class 2 belts shall have a making coat of glue (see 3.3.2) applied to the backing, and after the abrasive grain is deposited on the making coat, a sizing coat of a suitable nonhygroscopic synthetic resin shall be applied to complete the bond.

3.6.6 Class 3, resin bond. Class 3 belts shall have a making coat of a suitable nonhygroscopic synthetic resin (see 3.3.3) applied to the backing, and after the abrasive grain is deposited on the making coat, a sizing coat of a suitable nonhygroscopic synthetic resin shall be applied to complete the bond.

3.6.7 Class 4, waterproof (wet sanding). Class 4 belts shall have a making coat of a suitable nonhygroscopic resin (see 3.3.3) applied to a waterproof type cloth backing, and after the abrasive grain is deposited on the making coat, a sizing coat of a suitable nonhygroscopic synthetic resin shall be applied to complete the bond.

3.7 Type II band. Type II bands shall consist of two layers of backing cloth, an outer layer of abrasive coated cloth spirally wound with the edges butted and an inner layer of backing cloth spirally wound with the edges butted. The butted edges of the two layers shall be positioned so that the joints are not superimposed and the two layers of cloth shall be joined with a suitable adhesive to prevent separation under normal working conditions.

3.7.1 Marking for identification. Due to size, bands shall be identified only on the carton (see 5.3). The abrasive grain and grit size shall conform to table II.

3.7.2 Bands shall, when conditioned for at least 24 hours at 70 degrees ± 2 degrees F. and 50 percent plus or minus 2 percent relative humidity, conform to the dimensions shown in table IV for the size specified (see 6.2).

TABLE IV. Type II bands dimensions

Inside Diameter	Width
Tolerance: ± 1/32 when less than 1" I.D. ± 1/16 when 1" I.D. or over	Tolerance: Under 6 inches: ± 1/16 6 inches and over: 40 grit size and finer: ± 1/16 36 grit size and coarser: ± 3/32
Inches:	Inches:
1/2	1/4, 3/8, 1/2
3/4	3/4
1	1/2, 3/4, 1, 1-1/2, 2
1-1/2	1-1/2
2	1-1/2, 2, 2-1/2, 3, 6
2-1/8	8
3	1-1/2, 2, 2-1/2, 3, 6, 7-5/8

3.7.3 Class 1, glue bond. Class 1 bands shall have bonding adhesive coats in conformance with 3.6.4.

3.7.4 Class 2, resin over glue bond. Class 2 bands shall have bonding adhesive coats in conformance with 3.6.5.

3.7.5 Class 3, resin bond. Class 3 bands shall have bonding adhesive coats in conformance with 3.6.6.

3.8 Workmanship. Workmanship shall be of the highest grade and equal in every respect to good commercial practice. Abrasive belts and bands shall be free from imperfections which may impair their durability or serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling.

4.2.1 Lot. All type I belts or type II bands of the same class, grit size, and composition presented at one time shall be considered a lot for purposes of examination and tests.

4.2.2 Sampling for quality conformance inspection. A random sample of belts or bands shall be selected from each lot in accordance with MIL-STD-105 at inspection level I. The acceptable quality level (AQL) shall be 2.5 percent defective.

4.3 Quality conformance inspection.

4.3.1 Visual and dimensional examination. Each of the sample belts or bands selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any belt or bands in the sample containing one or more major visual or dimensional defects shall not be offered for delivery, and if the number of defective belts or bands in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.3.2 Tests. Each of the sample belts or bands selected in accordance with 4.2.2 shall be subject to the test of 4.4. If any sample belt or band fails to pass the test, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 Grading for grit size. Grading to determine the grit size of abrasive grain including recovery of grain from the coated product, preparation of grain for grading and procedure shall be in accordance with PS-8.

4.4.2 Certification of grading. Unless otherwise specified (see 6.2), grading tests may be waived upon certification by the contractor that the grit size of the abrasive grain indicated on his product was graded in accordance with PS-8.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing. Belts or bands of like description, in quantities as specified (see 6.2) shall be packaged and packed in accordance with MIL-P-3816. The level of packaging shall be A, B, or C as specified (see 6.2) and the level of packing shall be A, B, or C as specified (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.2.1 or 5.2.2 (see 6.2).

5.2.1 Civil agencies. All marking shall be in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. All marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The belts and bands covered by this specification are intended for use on power tools for sanding and finishing of wood, metal, ceramics and synthetic plastic articles.

Use open coat compositions 2, 6, and 7 where material sanded is apt to load the abrasive coated surface.

J Weight garnet items, compositions 4 and 6, should be used where conformability is required in wood working.

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6.1.1 Type I, endless belts, are available in all classes and compositions and type II, bands, are available only in certain of the classes and compositions (see table II).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class and composition of belts or bands required (see 1.2.1).
(c) Grit size of belts and bands required (see 3.4 and table II).
(d) Size of belt required (see 3.6.3 and table III).
(e) Size of band required (diameter X width) (see 3.7.2 and table IV).
(f) Specify when certification of the grit size of the abrasive grain is not acceptable in lieu of tests (see 4.4.2).
(g) Level of packaging and packing required (see 5.1 and 5.2).
(h) Quantity of belts or bands required (see 5.1 and 5.2).

MILITARY INTERESTS:

Army - MR
Air Force - 84

User activities:

Navy - YD, SH
Army - ME

Preparing activity:

Air Force - 84

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Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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